

Understanding Middleware in LangChain and DeepAgents



Learning Goals

- What agents are
- What the agent loop is
- What middleware is in LangChain and why it exists architecturally
- How middleware interacts with the agent loop
- How to use built-in and custom middleware
- How DeepAgents use middleware to extend runtime behavior



Motivation



AI Finance Agent

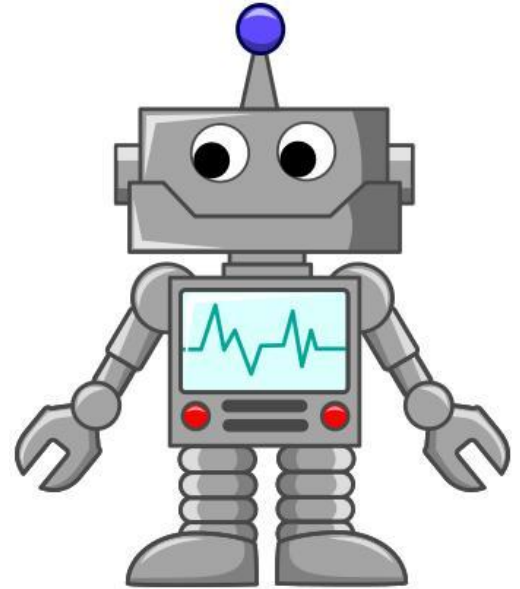
Scenario:

You are a developer at a financial company.

You decide to build an AI Finance Operations Assistant.

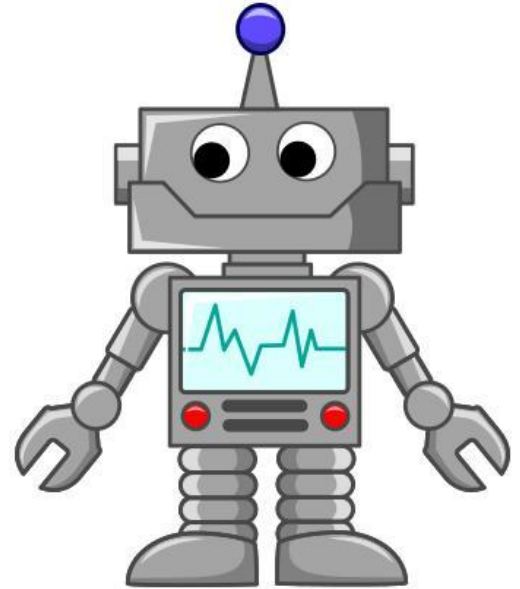
The assistant will:

- reimburse employees
- send payments
- access invoices
- send finance emails



AI Finance Agent

- How should the assistant log actions for auditing?
- How will it require approval for risky actions?
- How will it prevent unsafe tool usage?



What is an Agent?



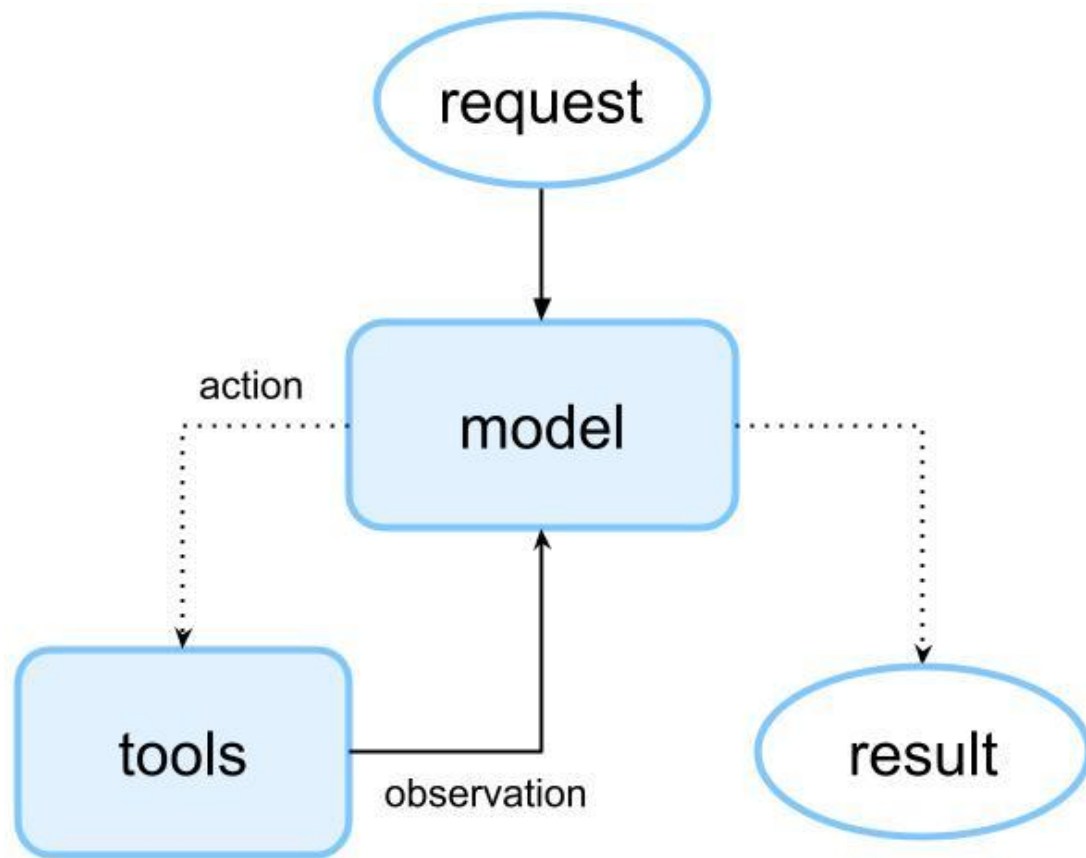
What is an Agent?

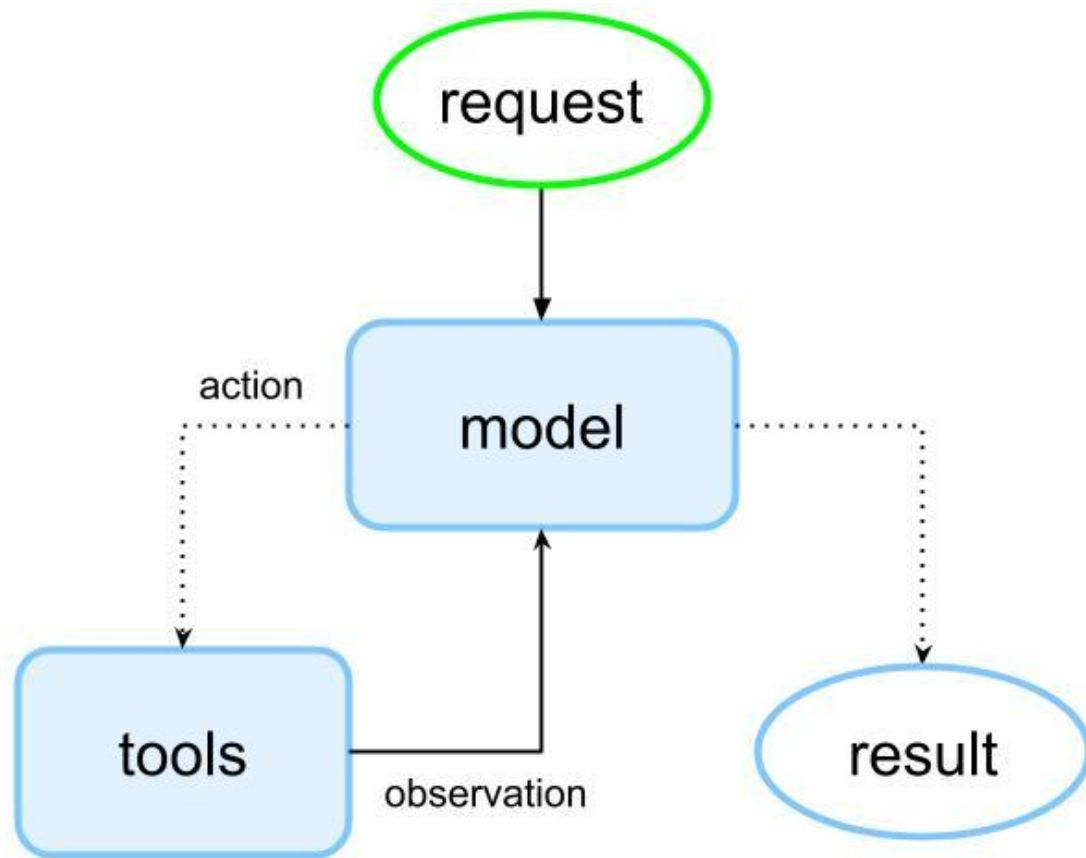
- Reason about a task
- Decide which tools to use
- Observe tool results
- Continue iterating until the task is complete

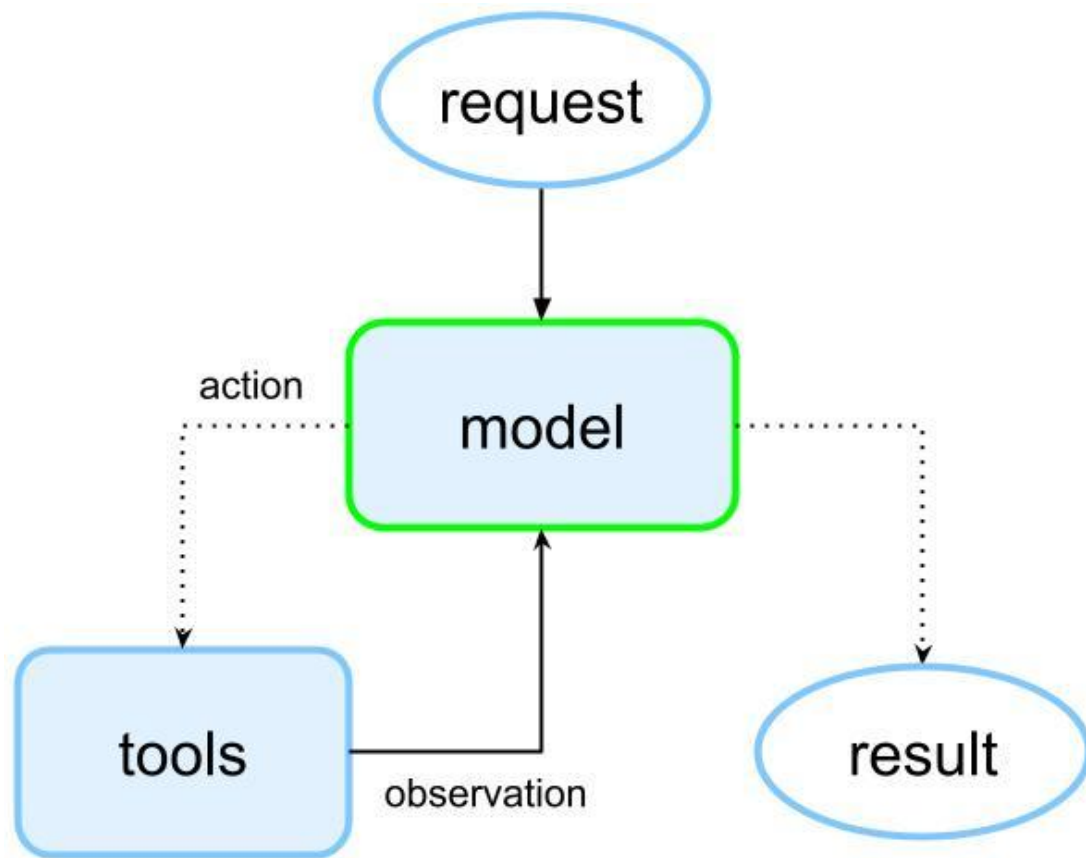


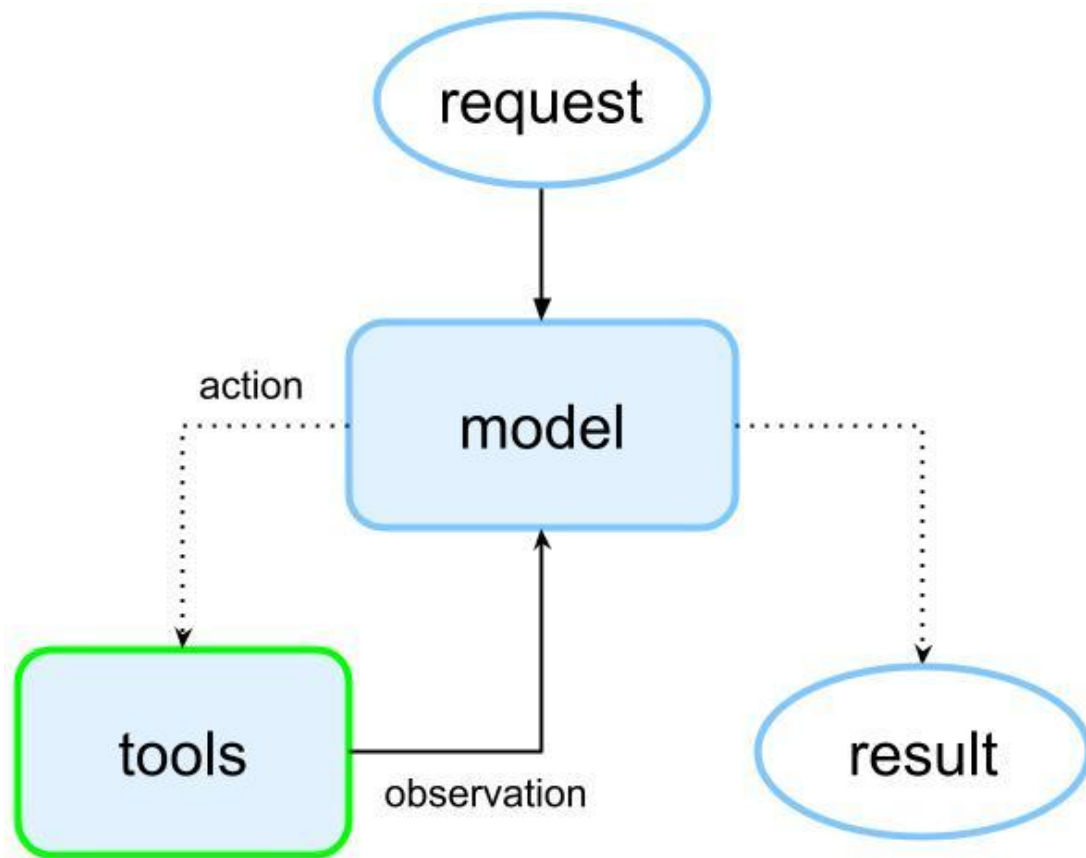
Agent Loop

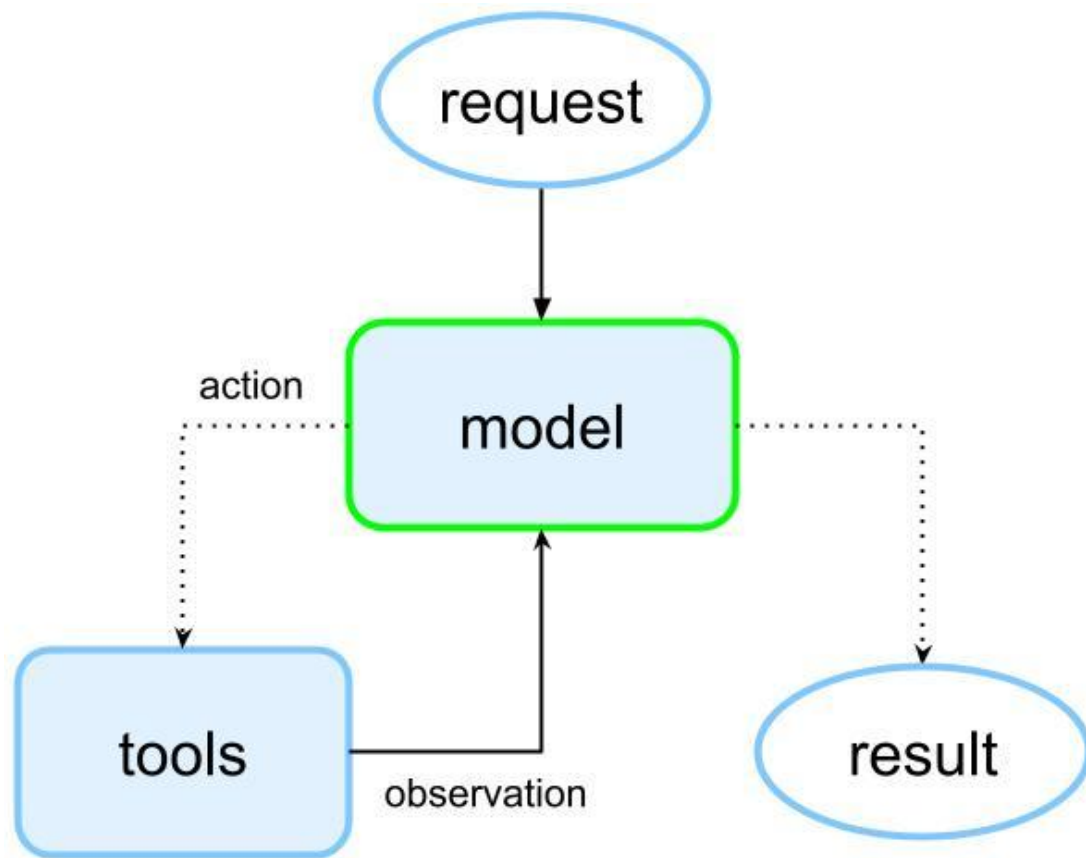


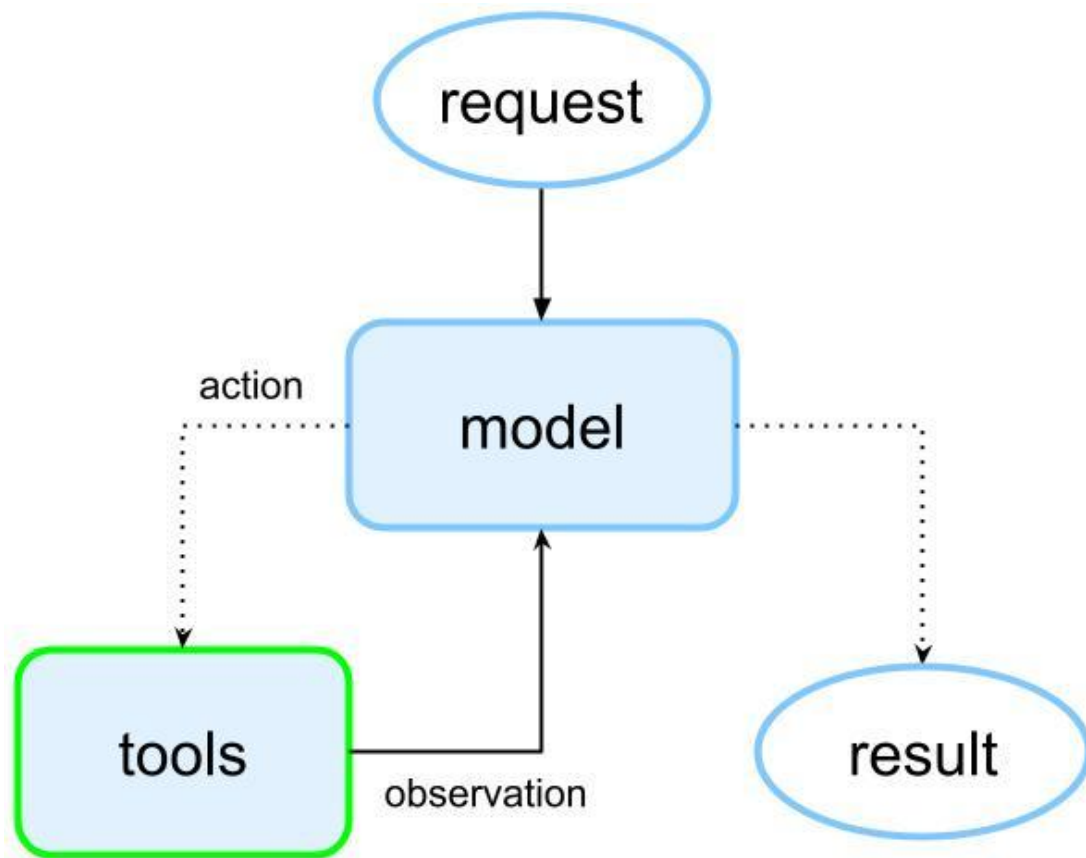


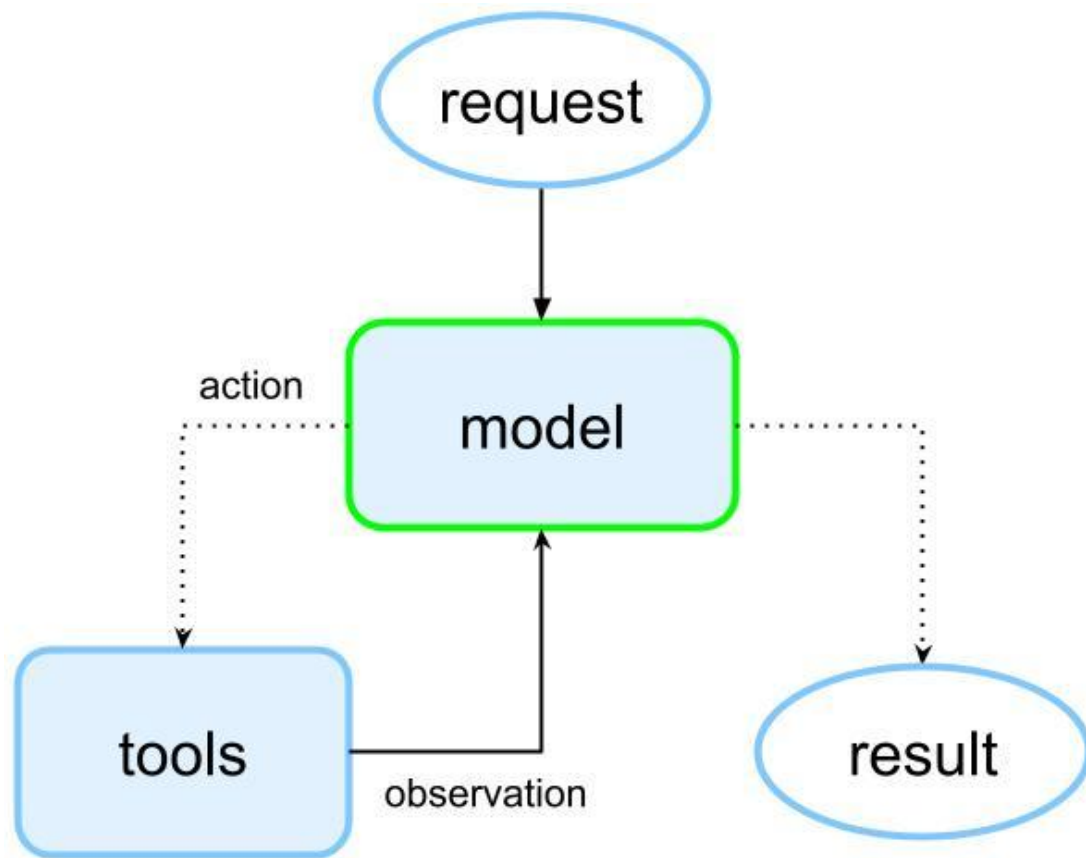


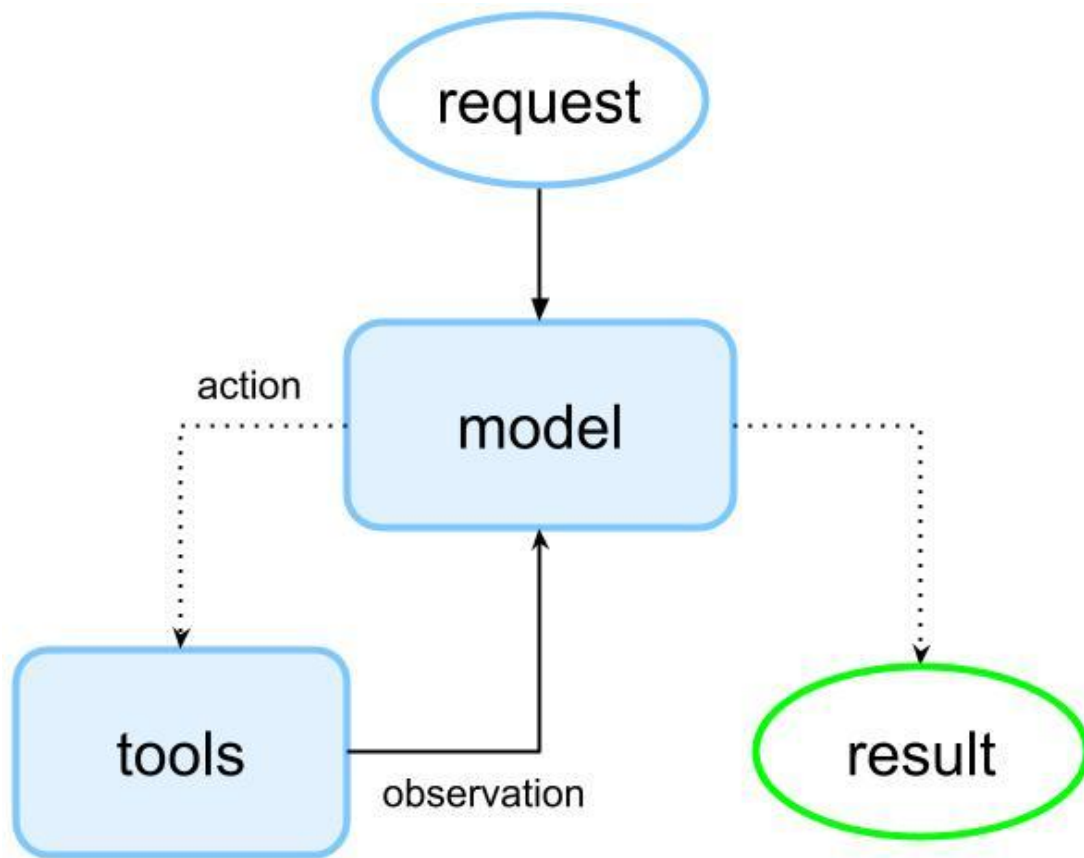












What is Middleware?



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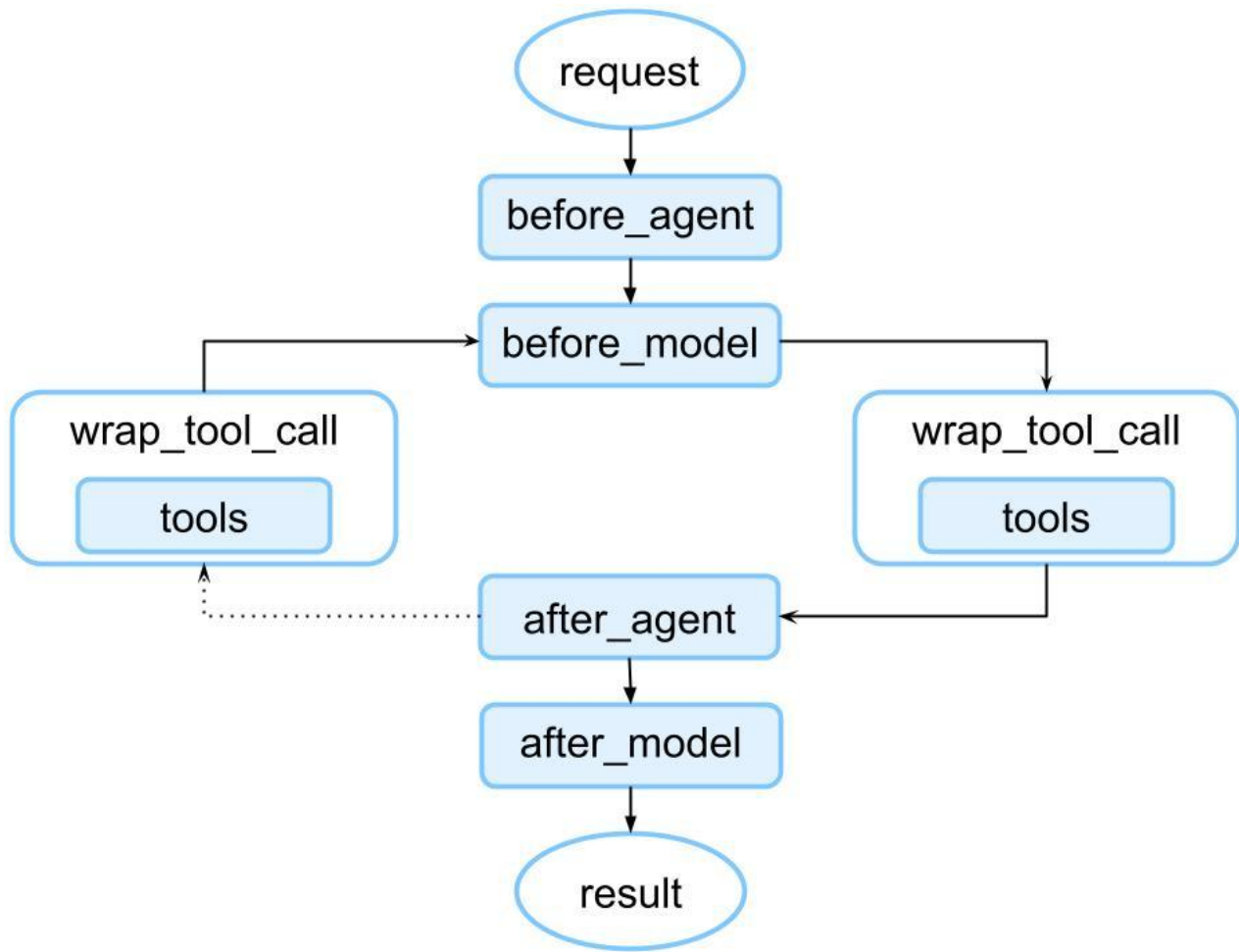
- Observe execution
- Modify execution
- Interrupt execution



What is Middleware Used For?

- Logging
- Approvals
- Retries
- Summarization
- Safety guardrails





Pre-built Middleware Examples



Custom Middleware Examples



DeepAgents Connection



DeepAgents

- More advanced runtime designed for long-running workflows, planning, memory, subagents, and durable execution
- Deep agents use middleware extensively.



Additional Notes



Custom Middleware

Decorator-Based Middleware

- Single-hook behaviors
 - ◆ Logging
 - ◆ Request modification

Class-Based Middleware

- Multi-hook behaviors
- Shared configuration
- Reusable state



Extending this Example

- Medical systems
- Customer support
- DevOps agents
- Autonomous coding agents

